

Hui Zhu

Computer Vision & Trustworthy AI Researcher

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[LinkedIn](#) · [Google Scholar](#)

Computer Science Ph.D. researcher building **multimodal vision systems and trustworthy AI methods**. Published at WACV and ECAI, with hands-on experience in 2D/3D segmentation, vision-language fusion, foundation-model adaptation, and LLM privacy evaluation. Skilled at taking research from data pipeline and experiment design through multi-GPU evaluation and publication.

Focus	Specialty	Methods	Practice
Computer Vision	Multimodal Learning	Foundation Models	Reproducible ML

EDUCATION

Ph.D. in Computer Science · Wayne State University Expected Jul 2028
Detroit, Michigan · Advisor: Prof. Dongxiao Zhu

B.Eng. in Software Engineering · Dalian University of Technology Jul 2023
Dalian, China

RESEARCH EXPERIENCE

- Graduate Research Assistant · Trustworthy AI Lab, Wayne State University** Aug 2023—Present
- Develop and evaluate segmentation, vision-language, and trustworthy-AI methods across clinical, microscopy, geographic, and language-model data.
 - Co-developed **BiPVL-Seg**, combining bidirectional progressive fusion with global-local contrastive alignment for end-to-end vision-language segmentation.
 - Co-authored **MulModSeg** (WACV 2025), enabling one model to learn from unpaired modalities through text conditioning and alternating training without added inference cost.
 - Build reproducible PyTorch/MONAI pipelines for preprocessing, augmentation, multi-GPU training, ablation studies, calibration, and model evaluation.
 - Collaborate across computer science, engineering, and applied research teams; translate domain problems into measurable machine-learning objectives.
- Undergraduate Researcher · Wang Laboratory, Dalian University of Technology** Sep 2022—Jul 2023
- Applied deep-learning methods to experimental signal data, covering preprocessing, feature extraction, algorithm replication, and parameter studies.
 - Coordinated reruns and result analysis to improve experimental reliability and communicate progress to the research team.

Hui Zhu · Curriculum Vitae · 1 / 2

PUBLICATIONS & PREPRINTS · NEWEST FIRST

- Automatic Calibration for Membership Inference Attack on Large Language Models.**
Saleh Zare Zade, Yao Qiang, Xiangyu Zhou, **Hui Zhu**, Mohammad Amin Roshani, Prashant Khanduri, Dongxiao Zhu. **ECAI 2025**.
[arXiv:2505.03392](https://arxiv.org/abs/2505.03392)
- BiPVL-Seg: Bidirectional Progressive Vision-Language Fusion with Global-Local Alignment for Medical Image Segmentation.**
Rafi Ibn Sultan, **Hui Zhu**, Chengyin Li, Dongxiao Zhu. [arXiv preprint, Mar 2025](#). [arXiv:2503.23534](https://arxiv.org/abs/2503.23534)
- Mechanical Evolution of Metastatic Cancer Cells in 3D Microenvironment.**
Karlin Hilai, Daniil Grubich, Marcus Akrawi, **Hui Zhu**, Razanne Zaghloul, Chenjun Shi, Man Do, Dongxiao Zhu, et al. **Small**, Mar 2025.
[doi:10.1002/sml.202403242](https://doi.org/10.1002/sml.202403242)
- Longitudinal Mechanical Imaging of Metastatic Tumor Spheroids Using Brillouin Microscopy.**

Karlin Hilai, Daniil Grubich, Marcus Akrawi, **Hui Zhu**, Razanne Zaghloul, Chenjun Shi, Man Do, Dongxiao Zhu, Jitao Zhang. [Optical Elastography and Tissue Biomechanics XII, Mar 2025](#). doi:10.1117/12.3043385

5. **MulModSeg: Enhancing Unpaired Multi-Modal Medical Image Segmentation with Modality-Conditioned Text Embedding and Alternating Training.**

Chengyin Li, **Hui Zhu**, Rafi Ibn Sultan, Hassan Bagher Ebadian, Prashant Khanduri, Chetty Indrin, Kundan Thind, Dongxiao Zhu. [WACV 2025, pp. 3581–3591, Feb 2025](#). [paper](#)

6. **GeoSAM: Fine-Tuning SAM with Multi-Modal Prompts for Mobility Infrastructure Segmentation.**

Rafi Ibn Sultan, Chengyin Li, **Hui Zhu**, Prashant Khanduri, Marco Brocanelli, Dongxiao Zhu. [arXiv preprint, Nov 2023](#). [arXiv:2311.11319](#)

TEACHING & LEADERSHIP

Jan 2024—Present

Teaching Assistant / Project Instructor · CSC 4996 Capstone Projects

Wayne State University

- Guide student teams through problem definition, software architecture, implementation, milestone reviews, and final presentations.
- Supported projects spanning image analysis, compliance auditing, generative audio, voice interfaces, autonomous systems, and mobile applications.

May 2022

Algorithm Analysis Team Lead

- Coordinated literature review, code reproduction, parameter studies, experiment monitoring, and team progress.

TECHNICAL TOOLKIT

Programming Python, C/C++, Bash, SQL, LaTeX

ML & data PyTorch, MONAI, Hugging Face Transformers, scikit-learn, NumPy, SciPy, pandas

Vision U-Net, ResNet/CNNs, Vision Transformers, Swin-UNETR, SAM fine-tuning, vision-language and multimodal learning

Imaging 2D/3D segmentation, CT/MR and microscopy workflows, DICOM/NIfTI, OpenCV, scikit-image

Evaluation Dice, IoU, boundary metrics, ROC/AUC, cross-validation, ablation design, calibration, uncertainty estimation

Engineering Linux, Git/GitHub, Docker, CUDA, multi-GPU training, HPC scheduling, Conda, reproducible experiment tracking